



FEEDBACK

YOUR SUGGESTIONS

GOOD WISHES FROM A REGULAR READER

Thank you very much for remembering me even after my subscription expired! I am an amateur electronics hobbyist with interest in electronics since I was 16 years old. Now I am 62. Throughout these years, I have been a reader of EFY. Sometimes I read EFY in libraries, sometimes I purchase it from magazine stalls or subscribe to it. I used to assemble DIY projects. Nowadays DIY projects are based on high-end technology. I have all the good wishes for my alma mater in electronics.

Purnachandra Rath

Through e-mail

DIGITAL TRANSMITTERS PLEASE

I am a subscriber to EFY and work for Prasar Bharati. In our country where analogue radio and terrestrial TV transmissions still exist, a suitable option is required to convert the same to digital. I request EFY to publish a series of articles on viable options and encourage manufacturers to convince the government to introduce digital transmission systems. I am sure your contribution will work as a catalyst.

R.S. Evani

Through e-mail

AUDIO AMPLIFIER

This is regarding '30W Audio Power Amplifier using NE5534 and Darlington Transistors' DIY project published in November 2017 issue. NE5534 IC available in the local market is made in China, which may blow up any time. Similarly, almost all ICs used in audio amplifier circuits published by EFY, which are available in the local market, are made in China. These ICs are incapable of sustaining 35-40V or higher voltages. What should be the maximum safe operating voltages for these circuits?

Also, where can I get original ICs like TDA2030 and LM1875 used in the circuits published by EFY?

Anirvan Kule

Kolkata

From electronicsforu.com

Electronics Projects

I am interested in '1kW Sine-wave Inverter' DIY article published on your website <https://electronicsforu.com/electronics-projects/1kw-sine-wave-inverter-circuit>. Can I use the same circuit for 12V but with a different Ah-rated battery?

Amit

The author Dr R.V. Dhekalé replies: Yes, you can use 12V, 24V or 48V batteries to get a higher power output. You have to be careful while providing 12V tapping for the circuit. Also, you have to change the inverter transformer specifications accordingly.

In the above inverter circuit, how can I change the output frequency of the inverter to 240V, 60Hz (instead of 50Hz) for use in Québec, Canada?

Sylvain B.

The author Dr R.V. Dhekalé replies: The frequency can be changed from 50Hz to 60Hz by using the preset provided.

The 'Arduino Based GSM Home Security System' published on your website <https://electronicsforu.com/electronics-projects/arduino-gsm-home-security-system> is a nice and easy-to-design DIY project. Where can I find its source code?

Also, I could not convert the PCB in PDF to Gerber file. Please help.

John Lagos

The author Joy Mukherji replies: The source code is available on EFY website: <http://efymag.com/admin/issuepdf/Arduino%20based%20GSM%20home%20security%20system.rar>. This link is also provided at the end of the article. You can construct the circuit on a general-purpose PCB.

The author Petre Petrov replies:

Many ICs are produced in Asia (China, Taiwan, Malaysia, etc.). Most of these ICs correspond well to the information published in datasheets.

Consequently, in all cases, you should refer datasheets and application notes published by the manufacturers. If you find that a component or equipment is not as per the published technical parameters, you can contact its manufacturer and ask for compensations according to the local law!

TDA2030, LM1875 and NE5534/A are very popular and low-priced ICs, and these behave well according to datasheets. I did not find significant differences between

these ICs from different manufacturers.

FAN SPEED REGULATOR

The 'Fan Speed Regulator Using PIC16F73' DIY project published in August 2017 issue is quite useful. Regarding this project, I have certain doubts:

1. What is the delay time between transistors T1 and T2?
2. Is there power loss during this time?
3. Do we need to remove the start capacitor?
4. Can the circuit be modified for:
(i) Fast over-current detection to cut off T1 and T2 in case of cross conduction?
(ii) Use of IGBT for higher power?

Joey Dimaculangan

Through e-mail

The author Prof. K. Padmanabhan replies: The time between the conduction of the two transistors can be seen in the oscilloscope itself (shown in the article). At about 2ms, it is sufficient to prevent shorting. When there is no current flow, there will be no power loss. The starting capacitor of the fan motor need not be removed.

As regards modification, fast over-current detection will make the circuit complicated. It is not required for a simple fan drive. Higher-power application is not possible without more complex drive circuitry. To make the circuit simple, optoisolator H11D1 is used.

4x4x4 LED CUBE

I want to assemble the '4x4x4-LED Cube Using AT89C2051' DIY project published in October 2017 issue. I have seen similar projects using shift registers, but this one is much simpler. What changes are required in the circuit to make it an 8x8x8 LED matrix?

Christo George

Through e-mail

The author Pamarthi Kanakaraja replies: An 8x8x8 LED cube is somewhat complex. A similar project using 512 LEDs is available on the following web link:

<http://www.instructables.com/id/Led-Cube-8x8x8>